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Powerful-IoU: More straightforward and faster bounding box regression loss with a nonmonotonic focusing mechanism

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ABSTRACT

Bounding box regression (BBR) is one of the core tasks in object detection, and the BBR loss function significantly impacts its performance. However, we have observed that existing IoU-based loss functions suffer from unreasonable penalty factors, leading to anchor boxes expanding during regression and significantly slowing down convergence. To address this issue, we intensively analyzed the reasons for anchor box enlargement. In response, we propose a Powerful-IoU (PIoU) loss function, which combines a target size-adaptive penalty factor and a gradient-adjusting function based on anchor box quality. The PIoU loss guides anchor boxes to regress along efficient paths, resulting in faster convergence than existing IoU-based losses. Additionally, we investigate the focusing mechanism and introduce a non-monotonic attention layer that was combined with PIoU to obtain a new loss function PIoU v2. PIoU v2 loss enhances the capability to focus on anchor boxes of medium quality. By incorporating PIoU v2 into popular object detectors such as YOLOv8 and DINO, we achieved an increase in average precision (AP) and improved performance compared to their original loss functions on the MS COCO and PASCAL VOC datasets, thus validating the effectiveness of our proposed improvement strategies.

1. Introduction

Object detection is a crucial task in computer vision and has now been widely used in various application scenarios (Zou, Chen, Shi, Guo, & Ye, 2023), such as video surveillance (Chen, Yu, Kong, Wu, & Wen, 2020; Kang et al., 2017; Lee, Lee, Cho, & Kang, 2020; Liu, Liao, & Hu, 2019; Shao, Wang, Wang, Du, & Wu, 2019; Sun, Chen, Chao, Ruan, & Mukherjee, 2020), autonomous driving (Azevedo & Santos, 2022; Fan, Huo, & Li, 2020; Fanthony, Husin, Hikmarika, Dwijayanti, & Suprpto, 2021; Li, Ji, Qu, Zhou, & Cao, 2022; Liang et al., 2022; Qiu, Lu, Wang, & Jiang, 2023; Sarda, Dixit, & Bhan, 2021; Wu, Iandola, Jin, & Keutzer, 2017), robot vision (Girshick, 2015; Redmon, Divvala, Girshick, & Farhadi, 2016; Tang, Zhou, Wang, & Zhang, 2023), agricultural monitoring (Gevorgyan, 2022; Rezaatofghi et al., 2019; Yu, Jiang, Wang, Cao, & Huang, 2016; Zhang, Ren et al., 2022; Zheng et al., 2020), reverse engineering of integrated circuits (IC), etc. Object localization, a key task in object detection, heavily relies on the evaluation and optimization of the bounding box regression (BBR) loss function. Hence, the bounding box regression loss function significantly

impacts the performance of object detectors. Most BBR losses can be classified as l_n -norm and IoU-based losses. The l_n -norm losses can be summarized as follows:

$$Loss = F(\Delta t_1, \Delta t_2, \Delta t_3, \Delta t_4), \quad (1)$$

where Δt_1 , Δt_2 , Δt_3 , and Δt_4 represent the differences between the regression parameters ($xywh$ or $x_1y_1x_2y_2$) for the predicted box and target box, and F is a function. These losses are sensitive to changes in the target size. To mitigate the effect of target size, YOLOv1 (Redmon et al., 2016) proposed predicting the square root of the width and height of the bounding box. Fast R-CNN (Girshick, 2015) proposed l_1 -smoothing loss to be more robust to outliers. However, these improvements bring slight performance improvement, and the l_n -norm loss ignores the correlations in BBR variables ($xywh$) (Yu et al., 2016). To address these issues, IoU (Intersection over Union) loss (Yu et al., 2016) is proposed:

$$L_{IoU} = 1 - \frac{I}{U}, 0 \leq L_{IoU} \leq 1, \quad (2)$$

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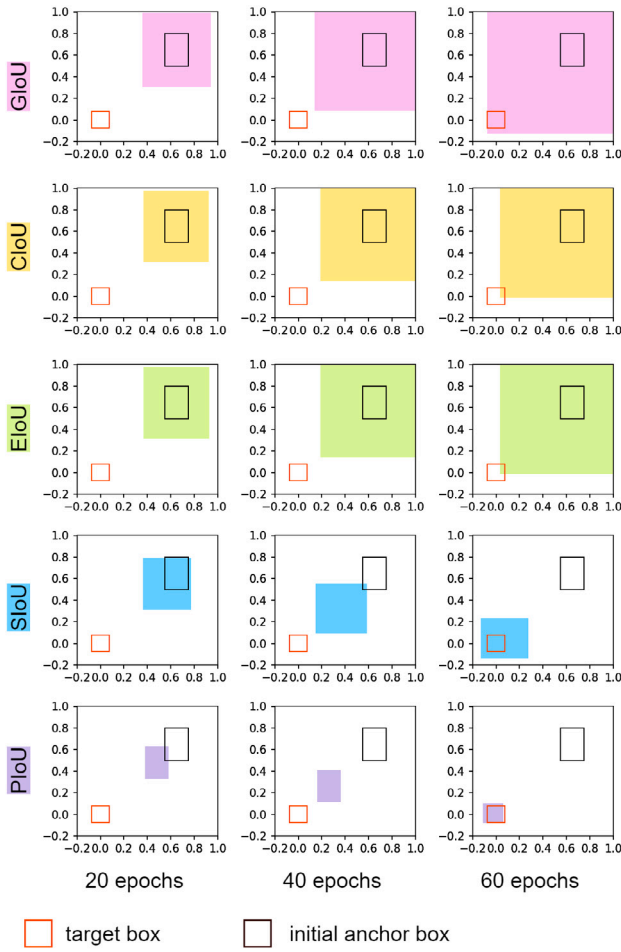


Fig. 1. Anchor box regression process guided by different IoU-based loss functions. The colored boxes are the anchor boxes guided by different loss functions during regression. It is clear that the anchor box guided by PIoU loss is the fastest to regress to approximate the target box. Moreover, the anchor box guided by all loss functions except PIoU loss suffers from the problem of area enlargement, while the anchor box guided by PIoU loss does not have this problem.

where I represents the intersection of the predicted box and the target box, and U represents their union. L_{IoU} treats the four boundaries of the anchor box as a whole for regression, effectively addressing the correlation issue. However, the gradient of L_{IoU} vanishes when there is no overlap between the anchor box and the target box.

Many IoU-based losses (Gevorgyan, 2022; Rezatofighi et al., 2019; Zhang, Ren et al., 2022; Zheng et al., 2020) have been proposed to address this limitation. Most of the IoU-based loss functions can be summarized as follows:

$$Loss = L_{IoU} + R(a(B, B_{gt}), b(B, B_{gt}), c(B, B_{gt}), \dots), \quad (3)$$

where B and B_{gt} represent the anchor box and the target box, respectively, and a, b, c , etc., are penalty factors. R is the penalty term, which is a function of the penalty factor as the independent variable. The penalty factors are typically geometric measures that quantify the matching degree between the anchor box and the target box.

However, the existing IoU-based loss functions still have unreasonable penalty factors. We evaluated the performance of different bounding box positions through simulated experiments. As shown in Fig. 1, the existing IoU-based loss functions initially increase the size of the anchor box during regression to achieve overlap with the target box, even when the area of the anchor box is already larger than that of the target box. This way of regression is complex and sluggish, requiring more epochs to converge. Furthermore, their penalty terms contain

unreasonable components that do not accurately reflect the discrepancy between the anchor box and the target box. They do not fully consider the target size or may exhibit degradation in certain cases.

To address the issues of anchor box enlargement and others, we propose a penalty factor with size adaptability that guides anchor boxes to regress directly and efficiently. We combine this penalty factor with a function that adapts the gradient based on the anchor box quality, resulting in a new IoU-based loss called Powerful-IoU (PIoU) loss. The PLoU loss directly minimizes the distances between the four edges of the anchor box and the corresponding edges of the target box. As shown in Fig. 1, during the early stages of regression, it guides the anchor box to move towards the target box in an almost straight path, leading to much faster convergence compared to existing IoU-based loss functions. While other IoU-based loss functions require at least 80–300 epochs to converge, the PLoU loss achieves alignment between the anchor box and the target box in just 60 epochs.

Furthermore, we explore the focusing mechanism (FM) in the BBR loss function. Focal EIoU (Zhang, Ren et al., 2022) introduces a regression version of the focal loss (Lin, Goyal, Girshick, He, & Dollár, 2017) called Focal EIoU, but its attention function is monotonic and does not fully exploit the potential of the attention mechanism. WIoU (Tong, Chen, Xu, & Yu, 2023) introduces a dynamic non-monotonic FM that evaluates the quality of individual anchor boxes based on the average quality of anchor boxes in the batch. However, WIoU relies on two hyperparameters, making it challenging to determine appropriate values for different datasets.

To address these issues, we design a non-monotonic attention layer controlled by a single hyperparameter and combine it with the PLoU loss, resulting in the PLoU v2 loss. This enhances the ability of PLoU to focus on anchor boxes of medium quality. By integrating PLoU v2 into popular object detectors such as YOLOv8 and DINO, we achieve SOTA detection accuracy compared to other loss functions.

The contributions of this paper can be summarized as follows:

1. We analyze the reasons behind anchor box enlargement in IoU-based loss functions and investigate the penalty factors and functional forms of the penalty term.
2. We propose the PLoU loss, which effectively addresses anchor box enlargement by incorporating penalty factors with desirable characteristics and a gradient-adjusting function based on anchor box quality. The PLoU loss demonstrates faster convergence compared to other IoU-based losses.
3. We design the PLoU v2 loss, which combines a non-monotonic attention layer controlled by a single hyperparameter. This enhances the focusing ability of PLoU on anchor boxes of medium quality.
4. By integrating PLoU v2 into popular object detectors such as YOLOv8 and DINO, we achieve the SOTA detection accuracy among all the loss functions, as demonstrated by object detection experiments.

2. Related work

2.1. BBR loss functions

An efficient loss function is crucial for achieving good performance in the localization task of object detectors. Over the years, many research works have been conducted in this area. To address the sensitivity of ln-norm loss functions to scale changes, YOLOv1 (Redmon et al., 2016) proposed predicting the square root of the width and height of the bounding box. Fast R-CNN (Girshick, 2015) introduced the l_1 -smooth loss to handle outliers more robustly. However, these l_n -norm loss functions assume independence among the four variables, which does not align with reality and leads to less accurate localization. To tackle this issue, IoU loss (Yu et al., 2016) is proposed. However, the gradient of IoU loss vanishes when there is no overlap between the two boxes, so IoU-based loss functions such as Generalized IoU (GIoU) (Rezatofighi et al., 2019) have been proposed.

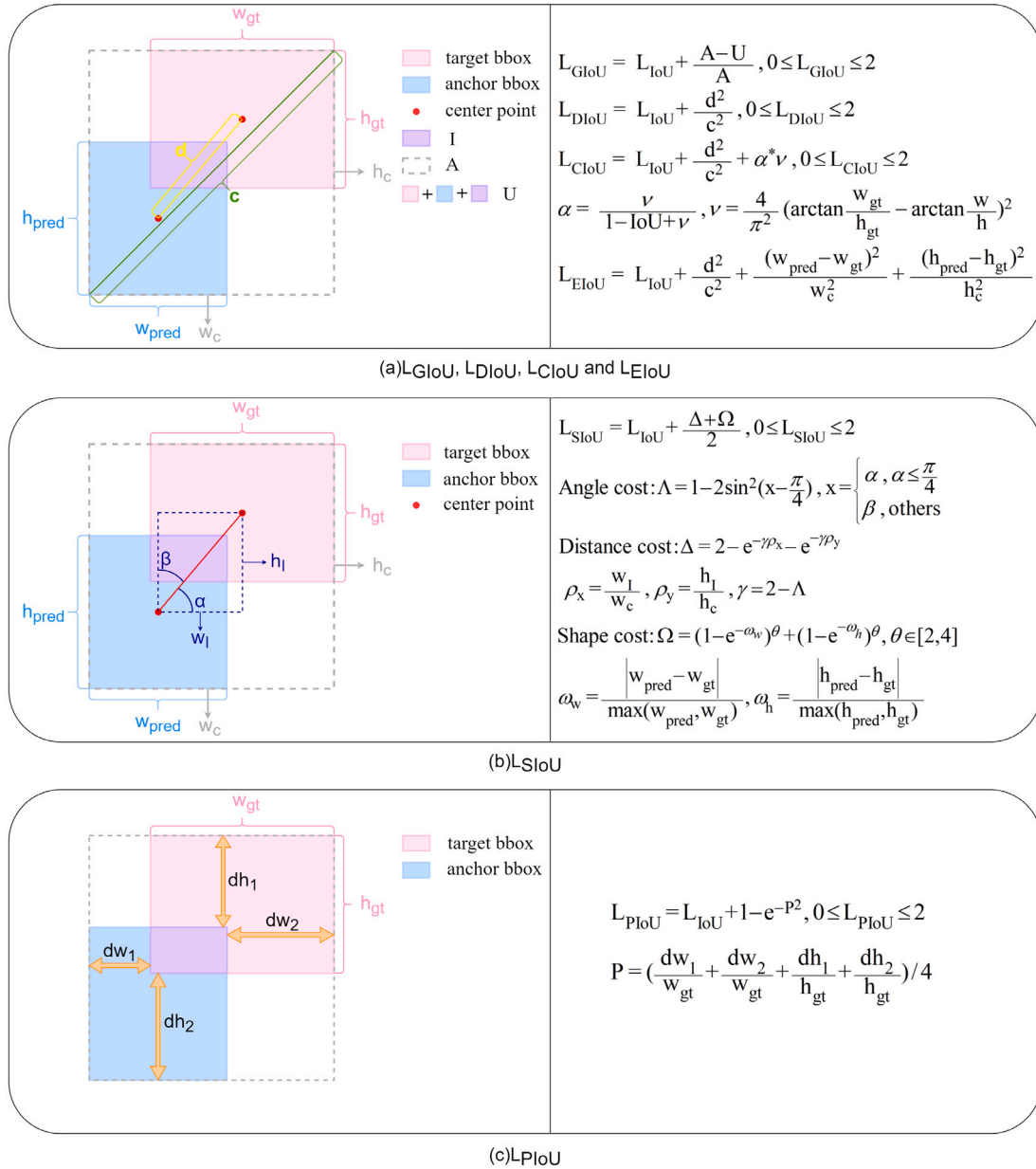


Fig. 2. IoU-based losses. The loss functions in both (a) and (b) use dimensional information such as the diagonal length or area of the smallest external bounding box (the gray dashed box) of the anchor box and the target box as the denominator in the loss factor. In contrast, the PIoU loss in (c) uses only the edge lengths of the target box as the denominator in the loss factor.

Many improvement works of IoU-based loss functions are carried out around the penalty term. Distance IoU (DIoU) (Zheng et al., 2020) introduced a penalty term that directly minimizes the normalized distance between the bounding boxes' center points, promoting the anchor box's movement towards the target box. Complete IoU (CIoU) (Zheng et al., 2020) built on DIoU by incorporating an aspect ratio penalty term, leading to faster convergence and improved performance. EIou (Zhang, Ren et al., 2022) replaced the penalty term of aspect ratio in CIoU with a more effective penalty term, aiming to directly reflect the differences in width and height between the anchor box and the target box. SIoU (SCYLLA-IoU) (Gevorgyan, 2022) designed penalty terms that consider three aspects: angle, shape, and distance. The formulas and diagrams of these loss functions are illustrated in Fig. 2 (a) and (b).

However, existing IoU-based loss functions still suffer from unreasonable penalty factors, resulting in anchor box enlargement during the regression process and slow convergence. While CIoU (Zheng et al.,

2020) and EIou (Zhang, Ren et al., 2022) acknowledged the issue of anchor box enlargement, they did not identify the problem's root cause or address it. Additionally, existing IoU-based losses exhibit limitations such as inadequate reflection of the discrepancy between the anchor box and the target box, insufficient consideration of target size, or degradation in certain cases.

To overcome these limitations, we propose PIoU, which combines a penalty factor with target box size as the denominator and a function that adapts to anchor box quality. This approach guides anchor boxes to regress efficiently along a more direct path, leading to faster convergence and higher accuracy.

2.2. Focusing mechanism in BBR loss functions

In BBR, there is an imbalance of training samples, i.e., the number of high-quality instances is much less than the number of low-quality instances. Recent works (Tong et al., 2023; Zhang, Ren et al., 2022)

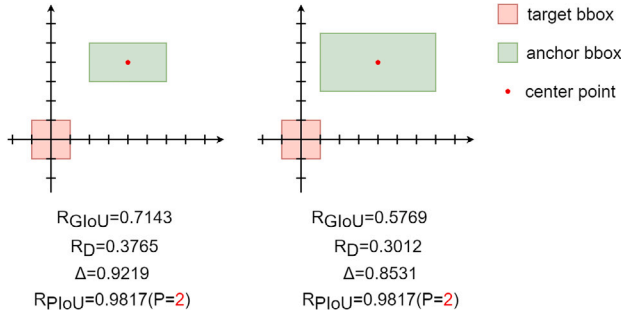


Fig. 3. An example of an expanded anchor box. In the left image, the center of the anchor box is located at (4, 4) with a width of 4 and a height of 2. In the right image, the center of the anchor box is also at (4, 4) but with a width of 6 and a height of 3. It can be observed that despite the increase in the size of the anchor box, R_{GIoU} , R_D , and Δ decrease. However, R_{PIoU} does not exhibit such behavior.

use a focusing mechanism to enable high-quality samples to contribute more gradients in model optimization.

EIoU (Zhang, Ren et al., 2022) proposed a regression version of the focal loss (Lin et al., 2017) called Focal EIoU to focus on high-quality examples:

$$L_{Focal-EIoU} = IoU^\tau L_{EIoU}, \quad (4)$$

where τ is a hyperparameter greater than 0. The attention function IoU^τ adjusts the loss function's gradient, enabling higher-quality anchor boxes (larger IoU^τ values) to receive larger gradients. The degree of focusing can be controlled by adjusting IoU^τ through τ . This method focuses on higher-quality anchor boxes by amplifying the gradient when each anchor box approximates the target box. However, the attention function of Focal EIoU is monotonic and does not fully utilize the potential of the focusing mechanism.

To address this issue, WIoU (Tong et al., 2023) introduces a dynamic non-monotonic FM loss function that concentrates on anchor boxes with moderate quality and suppresses harmful gradients from low-quality examples:

$$L_{WIoU} = r \cdot L_{IoU} \cdot e^{\frac{d^2}{c^2}}, r = \frac{\phi}{\delta \epsilon^{\phi-\epsilon}}, \phi = \frac{L^*_{IoU}}{L_{IoU}}, \quad (5)$$

where r is the attention function of WIoU, and δ and ϵ are hyperparameters controlling the gradient of r . The operation $*$ denotes the detach operation, and $\overline{L_{IoU}}$ represents the average L_{IoU} value of all anchor boxes in a batch. WIoU evaluates the relative quality of an anchor box compared to the average quality of all anchor boxes in the batch, assigning more attention to anchor boxes with moderate quality. However, WIoU relies on two hyperparameters, making it challenging to determine suitable values for different datasets.

To address these limitations, we design a non-monotonic attention function controlled by a single hyperparameter, resulting in a new loss function called PIoU v2. This enhances the ability to focus on anchor boxes of moderate quality. PIoU v2 achieves state-of-the-art accuracy in advanced object detectors YOLOv8 (Jocher, Chaurasia, & Qiu, 2023) and DINO (Zhang, Li et al., 2022).

3. Powerful intersection over union loss

In this section, we first analyze the reasons for the enlargement of anchor box during the regression and identify the limitations of existing IoU-based loss functions. We then introduce the PIoU loss and discuss the impact of the function form.

3.1. Limitations of IoU-based losses

3.1.1. Reasons for anchor box enlargement

The enlargement of anchor box hinders the efficiency of anchor box regression. In this section, we analyze the reasons behind this enlargement.

GIoU Loss: The GIoU loss (Rezatofighi et al., 2019) is defined as shown in Fig. 2(a). The penalty term, R_{GIoU} , in the GIoU loss is $(A - U)/A$, where A is the area of the smallest external box for the anchor box and the target box, and U is the area of their union. The gradient of R_{GIoU} , w.r.t A and U , is calculated as follows:

$$\frac{\partial R_{GIoU}}{\partial A} = \frac{U}{A^2}, \quad (6)$$

$$\frac{\partial R_{GIoU}}{\partial U} = -\frac{1}{A}. \quad (7)$$

When U increases, R_{GIoU} decreases; when A increases, R_{GIoU} increases, and vice versa. From Eq. (6) and (7), we can derive:

$$\left| \frac{\partial R_{GIoU}}{\partial U} \right| \cdot \frac{U}{A} = \left| \frac{\partial R_{GIoU}}{\partial A} \right| \quad (8)$$

In cases where the anchor box and the target box do not overlap ($U < A$), we have:

$$\left| \frac{\partial R_{GIoU}}{\partial U} \right| > \left| \frac{\partial R_{GIoU}}{\partial A} \right| \quad (9)$$

This means that when A and U have the exact change in value, U has a greater impact on R_{GIoU} . Consequently, when the area of the anchor box increases, both A and U increase, leading to a decrease in R_{GIoU} . To illustrate this issue clearly, we provide a specific example in Fig. 3. In this example, while the center point of the anchor box remains fixed, enlarging the anchor box size results in a significant decrease in R_{GIoU} . The anchor box enlargement causing a decrease in R_{GIoU} is due to the use of A as the denominator. If a denominator factor independent of A is employed, such that Eq. (9) does not hold, anchor box enlargement can be avoided.

CIoU Loss and EIoU Loss: CIoU (Zheng et al., 2020) and EIoU losses (Zhang, Ren et al., 2022) employ the term $R_D = d^2/c^2$ in their penalty terms, as shown in Fig. 2(a). Here, d is the diagonal length of the intersection between the anchor box and the target box, and c is the diagonal length of the smallest external box for the anchor box and the target box. The gradient of R_D , w.r.t c , is calculated as follows:

$$\frac{\partial R_D}{\partial d} = \frac{2d}{c^2}. \quad (10)$$

When c increases, R_D decreases, and vice versa. Therefore, when the anchor box and the target box do not overlap, L_{IoU} remains unchanged, and simply increasing the size of the anchor box reduces the CIoU (Zheng et al., 2020) loss and EIoU (Zhang, Ren et al., 2022) loss. Similarly, we provide a specific example in Fig. 3 to illustrate this issue. In the example, while the anchor box does not move, enlarging its size decreases R_D . This demonstrates that using c as the denominator in the penalty term is unreasonable.

SIoU Loss: The SIoU loss (Gevorgyan, 2022) is defined as shown in Fig. 2(b). The gradient of the distance cost Δ , w.r.t w_c and h_c , is calculated as follows:

$$\frac{\partial \Delta}{\partial w_c} = -e^{-\gamma \frac{w_l}{w_c}} \cdot \gamma \cdot \frac{w_l}{w_c^2}, \quad (11)$$

$$\frac{\partial \Delta}{\partial w_l} = e^{-\gamma \frac{w_l}{w_c}} \cdot \gamma \cdot \frac{1}{w_c}. \quad (12)$$

Here w_c and h_c are the width and height of the smallest external box for the anchor box and the target box, and w_l and h_l are the width and height of their union, and γ is related to the angle cost. According to Eq. (11) and (12), when w_c and h_c increase, the distance cost Δ decreases. Similarly, we illustrate the change in the distance cost Δ with a specific example in Fig. 3. When the anchor box does not overlap with the target box, enlarging the anchor box area increases the area

of the smallest external box, resulting in a decrease in the distance cost Δ and the SIOU loss. Similarly, this indicates that using w_c and h_c as the denominator in the penalty term is incorrect.

Summarizing the reasons behind anchor box enlargement in these loss functions, we observe that using factors related to the size of the smallest external box for the anchor box and the target box as the denominator in the penalty term is unsuitable. This is the reason for the enlargement of anchor box areas during regression. As anchor box enlargement significantly hampers the efficiency of regression, IoU-based regression loss functions require more appropriate penalty terms. To address this issue, we propose a new loss function, PIoU, which incorporates a penalty term with the size of the target box as the denominator. PIoU avoids anchor box enlargement and adapts better to target sizes.

3.1.2. Limitations of IoU-based loss functions

This section analyzes the limitations of four IoU-based loss functions: GIoU (Rezatofighi et al., 2019), CIoU (Zheng et al., 2020), EIoU (Zhang, Ren et al., 2022), and SIOU (Gevorgyan, 2022) losses.

Limitations of GIoU Loss: As shown in Fig. 2(a), the penalty term of GIoU (Rezatofighi et al., 2019) loss is defined as $(A - U)/A$. When the anchor box and the target box overlap, $|A - U|$ is usually a small value. However, when a containment relationship exists between the anchor box and the target box, $A - U$ becomes 0, and the GIoU loss degenerates into the IoU loss. These result in slow convergence when the anchor box and the target box overlap in the later stages of regression.

Limitations of CIoU: As shown in Fig. 2(a), the penalty term of CIoU (Zheng et al., 2020) includes two terms that reflect the differences in the center point distance and aspect ratio. However, it does not directly capture the shape difference between the anchor box and the target box, leading to undesired convergence behavior (Zhang, Ren et al., 2022). Additionally, neither of the two terms in the penalty term reflects the changes in the size of the target box.

Limitations of EIoU: As shown in Fig. 2(a), the penalty term of EIoU (Zhang, Ren et al., 2022) considers both the center point distance and shape difference between the anchor box and the target box. However, during regression, the difference in size between the anchor box and the target box is much smaller than the size of their smallest external box. As a result, the two terms related to shape difference are usually small values, leading to small gradients that cannot prevent anchor box enlargement.

Limitations of SIOU: The penalty term of SIOU (Gevorgyan, 2022) loss considers three factors: angle, distance, and shape. However, SIOU loss is a complex loss function. Moreover, the distance cost does not consider the adaptability to the target size. Similarly, when the length or width of the anchor box is larger than that of the target box, the shape cost also overlooks the target size.

3.2. Proposed method

To address the issue of anchor box enlargement and the limitations of existing IoU-based loss functions, we propose a penalty factor P that adapts to the target size. The penalty factor is defined as follows:

$$P = \left(\frac{dw_1}{w_{gt}} + \frac{dw_2}{w_{gt}} + \frac{dh_1}{h_{gt}} + \frac{dh_2}{h_{gt}} \right) / 4. \quad (13)$$

Here, dw_1 , dw_2 , dh_1 , and dh_2 are the absolute values of the distances between the corresponding edges of the predicted box and the target box, and w_{gt} and h_{gt} represent the width and height of the target box, as illustrated in Fig. 2(c).

Using P as the penalty factor in the loss function does not cause anchor box enlargement, as confirmed by the experimental results in Fig. 1. This is because the denominator of P only depends on the size of the target box and is independent of the size of the smallest external box of the anchor box and the target box. Unlike the penalty factors used in other loss functions, anchor box enlargement does not change P , as

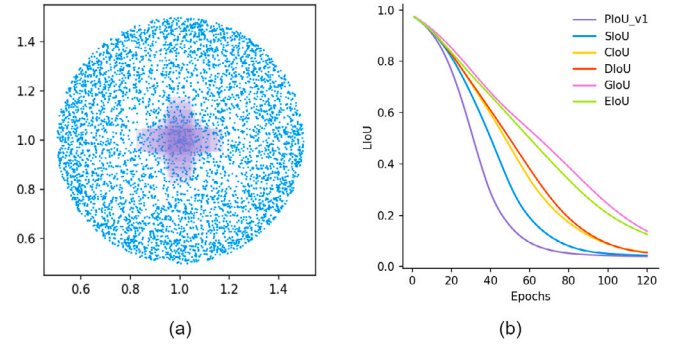


Fig. 4. Simulation experiments of multiple boxes regression: (a) 1,715,000 regression cases are adopted by considering different distances, scales and aspect ratios; (b) L_{IoU} of different loss functions at epochs.

demonstrated in Fig. 3. Furthermore, unless the anchor box completely overlaps with the target box, P will never degenerate to 0. Additionally, P exhibits adaptability to the target size.

To further improve the loss function, we propose a penalty function that adapts to the quality of the anchor box. The specific selection and study of these functions will be discussed in Section 3.3.

$$f(x) = 1 - e^{-x^2}, \quad (14)$$

$$PIoU = IoU - f(P), -1 \leq PIoU \leq 1, \quad (15)$$

$$L_{PIoU} = 1 - PIoU = L_{IoU} + f(P), 0 \leq L_{PIoU} \leq 2. \quad (16)$$

To differentiate from PIoU v2, the subsequent appearance of PIoU v1 refers to PIoU. The PIoU loss adapts to the target size and guides the anchor box to regress through a more direct and efficient path. To demonstrate the superiority of PIoU in terms of convergence speed, we conducted regression simulation experiments similar to those conducted in CIoU (Zheng et al., 2020). The experiments involved generating target boxes with an area of $1/32$ and 7 aspect ratios (1:4, 1:3, 1:2, 1:1, 2:1, 3:1, 4:1) at position (1, 1). We uniformly generated 5000 anchor points in a circular region centered at (1, 1) with a radius of 0.5. Additionally, we set 49 anchor boxes, including 7 scales ($1/32$, $1/24$, $3/64$, $1/16$, $1/12$, $3/32$, $1/8$) and 7 aspect ratios (1:4, 1:3, 1:2, 1:1, 2:1, 3:1, 4:1). In total, there were 1,715,000 ($7 \times 7 \times 7 \times 5,000$) regression cases, as shown in Fig. 4(a). The learning rate was set to 0.01, the optimizer used was Adam, and the total number of training epochs was set to 120, with L_{IoU} used as the evaluation metric.

As shown in Fig. 4(b), the experimental results demonstrate that our proposed PIoU achieves the fastest convergence speed among the compared methods.

3.3. Impact of function form

In this section, we investigate and propose suitable penalty functions for BBR and compare three different function forms. The function plays a crucial role in determining the gradient magnitude of the penalty term.

We refer to the desirable properties listed by EIoU (Zhang, Ren et al., 2022) for an ideal loss function, which include: (1) the gradient magnitude should approach zero as the regression error approaches zero, and (2) the gradient magnitude should increase rapidly for small regression errors and gradually decrease for large regression errors.

In addition to these properties, we believe a penalty function suitable for anchor box convergence should exhibit the following characteristics: extremely poor-quality or high-quality anchor boxes should have a small gradient; the maximum gradient should be at medium-quality anchor boxes. Poor-quality anchor boxes can introduce harmful

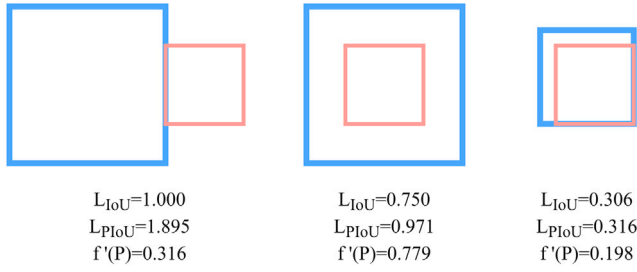


Fig. 5. Different quality of anchor boxes with their corresponding values of L_{IoU} , L_{PIoU} and gradient $f'(P)$. The red box is the target box, and the blue box is the anchor box. Target box is a square with a side length of 0.5, and anchor box is a square with side lengths of 1, 1, and 0.6, respectively. The three cases are a poor-quality anchor box, a medium-quality anchor box, and a high-quality anchor box.

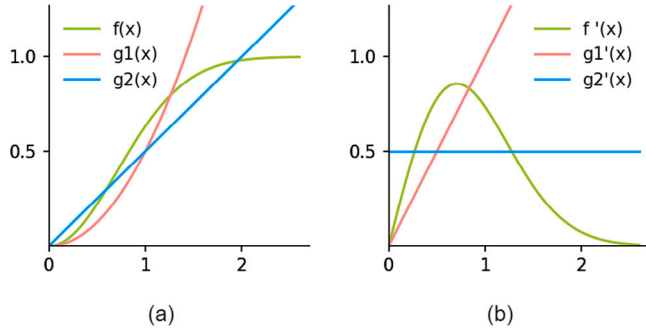


Fig. 6. Function image. (a) $f(x)$, $g_1(x)$ and $g_2(x)$. (b) The derivative of $f(x)$, $g_1(x)$ and $g_2(x)$.

gradients to the object detector (Zhang, Ren et al., 2022), so the penalty term should have a small gradient for poor-quality anchor boxes. The gradient should increase accordingly as the extremely poor-quality anchor boxes gradually regress. Medium-quality anchor boxes should have the maximum gradient, allowing them to regress quickly toward the vicinity of the target box and become high-quality anchor boxes. A smaller gradient should follow this to stabilize the optimization process toward complete alignment with the ground truth box. In this way, the object detector is able to focus on stages with anchor boxes that are of moderate quality (some poor-quality initial anchor boxes will become medium-quality after some epochs of training). Additionally, the function should have a fixed finite range of values for stability.

In Section 3.2, we propose a function $f(x)$ that satisfies all these characteristics. The function $f(x)$ is designed to adaptively adjust the gradient magnitude based on the quality of the anchor box. The specific function form of $f(x)$ and its corresponding gradient are illustrated in Fig. 6.

When the penalty factor P is large (>2), indicating a significant discrepancy between the anchor box and the target box, $f'(P)$ takes on a small value, which suppresses harmful gradients from low-quality anchor boxes. When P is around 1, indicating anchor boxes near the target box, the large $f'(P)$ leads to faster regression. As P approaches 0, indicating anchor boxes close to the target box, the $f'(P)$ gradually decreases with the improvement of anchor box quality, allowing the anchor boxes to optimize stably towards complete alignment with the target box. Examples of anchor boxes and their corresponding gradients in three different cases (poor quality, medium quality, and high quality) are shown in Fig. 5.

To demonstrate the superiority of our chosen function form and the importance of the penalty term's function form in the BBR loss function, we combine the same penalty term with different functions to form different loss functions.

$$g_1(x) = \frac{x^2}{2}, L_1 = L_{IoU} + g_1(P) \quad (17)$$

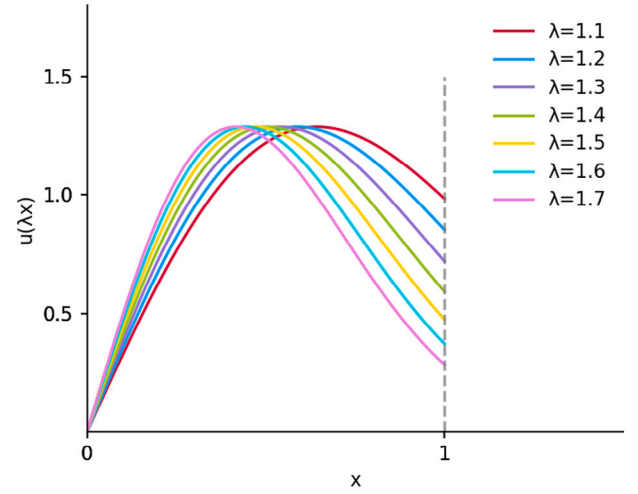


Fig. 7. $u(\lambda x)$ with different values of hyper-parameter λ .

$$g_2(x) = 0.5x, L_2 = L_{IoU} + g_2(P) \quad (18)$$

The functions $f(x)$, $g_1(x)$ and $g_2(x)$ and their gradients are illustrated in Fig. 6. We conduct ablation experiments by combining these functions with YOLOv8 using L_1 and L_2 losses. The detailed experimental results and analysis are presented in Section 5.3.

4. PIoU v2 with focal mechanism

In this section, we explore the application of the focusing mechanism to the loss function we proposed. To address the limitations of Focal EIoU (Zhang, Ren et al., 2022) and WIoU (Tong et al., 2023) mentioned in Section 2.2, we design a non-monotonic attention function combined with PIoU controlled by a single hyperparameter. PIoU v2 loss, obtained after adding an attention layer to PIoU loss, enhances the ability to focus on medium and high quality anchor boxes, boosting the performance of the object detector. The attention function and PIoU v2 loss are defined as:

$$q = e^{-P}, q \in (0, 1], \quad (19)$$

$$u(x) = 3x \cdot e^{-x^2}, \quad (20)$$

$$L_{PIoU, v2} = u(\lambda q) \cdot L_{PIoU} = 3 \cdot (\lambda q) \cdot e^{-(\lambda q)^2} \cdot L_{PIoU}. \quad (21)$$

Here, $u(\lambda q)$ represents the attention function, and its function plot is shown in Fig. 7. We replace the penalty factor P with q , which measures the quality of the anchor box, and q ranges from (0,1]. When $q = 1$, it implies $P = 0$, indicating complete alignment between the anchor box and the target box. As P increases, q gradually decreases, representing lower-quality anchor boxes. λ is the hyperparameter controlling the behavior of the attention function.

The non-monotonic attention function $u(\lambda q)$ enhances the ability of PIoU to focus on anchor boxes with moderate quality. Therefore, the PIoU v2 loss not only incorporates a non-monotonic attention function but also fully exploits the potential of the attention mechanism. By focusing on the stage of moderate quality during the regression process of each anchor box, PIoU v2 pays more attention to moderate-quality anchor boxes. PIoU v2 requires only a single hyperparameter, simplifying the tuning process.

By incorporating PIoU v2 into advanced object detectors such as YOLOv8 and DINO, we achieve SOTA detection accuracy compared to other loss functions. The detailed experimental results and analysis are presented in Section 5.2.

Table 1

The results of comparison experiments of YOLOv8 on the MS COCO dataset.

	AP75val	APval	AP50val
CloU (Zheng et al., 2020)	52.46	48.3	64.94
EIoU (Tong et al., 2023)	52.68	48.33	64.89
SiIoU (Gevorgyan, 2022)	52.78	48.37	64.97
GIoU (Rezatofighi et al., 2019)	52.84	48.3	64.77
PIoU v1	52.72	48.43	65.1
Focal-EIoU (Zhang, Ren et al., 2022)	51.78	47.29	63.68
WIoU (Tong et al., 2023)	52.32	47.98	64.44
PIoU v2	53.26	48.63	65.42

5. Experiments

5.1. Experimental setup

For fair comparisons, all our experiments were conducted on the PyTorch framework.

Dataset. For the comparison experiments of different loss functions, we used the MS-COCO (Lin et al., 2014) dataset and the VOC (Everingham, Van Gool, Williams, Winn, & Zisserman, 2010) dataset. For ablation experiments, we selected 20 categories from the MS-COCO dataset (Lin et al., 2014), consisting of 28,474 images for training and 1,219 images for validation. To evaluate the generalization of our proposed method in practical applications, we also created a dataset named the IC SEM dataset for integrated circuit reverse engineering using SEM (Scanning Electron Microscope) images of 55 nm integrated circuit chips. The dataset contains 80 images with 2,457 instances for training and 40 images with 1,332 instances for validation. The targets in this dataset include MCC (Multi-contact Cell), MOSFET (Metal Oxide Semiconductor Field Effect Transistor), and Filler.

Models. We selected the advanced object detectors YOLOv8 m (Jocher et al., 2023) and DINO-4scale (Zhang, Li et al., 2022). Except for the loss function and batch size, all other settings were kept at the default values of their source codes. For experiments of YOLOv8 on COCO datasets in Table 1, batch size is 4. For other experiments of YOLOv8, batchsize is 16. For experiments of DINO, batchsize is 32. All YOLOv8 experiments were trained on an NVIDIA 3090 GPU, and all DINO experiments were trained on an NVIDIA V100 GPU. When not specified, λ for PIoU v2 is set to 1.3 by default.

5.2. Comparison experiments

YOLOv8 on MS COCO dataset. We report the comparison results of different loss functions combined with YOLOv8 m (Jocher et al., 2023) on the COCO (Lin et al., 2014) dataset, as shown in Table 1. Among all the basic versions of loss functions (GIoU, CloU, SiIoU, EIoU and PIoU v1), our PIoU v1 achieved the best results. PIoU v2 achieved SOTA accuracy among all loss functions, with gains of 0.5% AP_{75} , 0.5% AP_{50} , and 0.3% AP . We observed that the performance improvement from other loss functions was marginal, while the improvement from PIoU v2 was relatively significant. Additionally, Focal-EIoU (Zhang, Ren et al., 2022) showed a performance decrease compared to EIoU (Zhang, Ren et al., 2022), indicating that the attention function of Focal-EIoU lacks generalization.

To assess the model's performance improvement, we ran the model on some images from COCO2017-val to compare the predicted results of anchor boxes. Fig. 8 shows some examples. It can be seen that the model using PIoU v2 is more sensitive to incomplete targets and can more accurately identify the class of targets by part of them, while the models using Focal-EIoU (Zhang, Ren et al., 2022) and WiIoU (Zhang, Ren et al., 2022) miss incomplete targets and have errors in their localization and classification.

DINO on PASCAL VOC dataset. We report the comparison results of DINO (Zhang, Li et al., 2022) combined with the focal mechanism loss function on the VOC (Everingham et al., 2010) dataset in

Table 2

The results of comparison experiments of DINO on the PASCAL VOC dataset.

	AP75val	APval
CloU (Zheng et al., 2020)	69.67	63.52
EIoU (Zhang, Ren et al., 2022)	69.76	63.65
SiIoU (Gevorgyan, 2022)	69.49	63.59
GIoU (Rezatofighi et al., 2019)	69.42	63.61
PIoU v1	69.66	63.52
Focal-EIoU (Zhang, Ren et al., 2022)	69.83	63.94
WIoU (Tong et al., 2023)	70.00	63.83
PIoU v2	70.50	64.23

Table 3

Performance of each BBR loss on IC SEM dataset.

	AP^{val}_{75}	AP^{val}_{50}
CloU (Zheng et al., 2020)	95.1	98.2
SiIoU (Gevorgyan, 2022)	95.4	98.7
EIoU (Zhang, Ren et al., 2022)	95.5	98.6
PIoU v1	95.7	98.7
Focal-EIoU (Zhang, Ren et al., 2022)	95.3	98.6
WIoU (Tong et al., 2023)	95.3	98.7
PIoU v2	96.0	98.9

Table 4Performance of PIoU v2 with different values of hyper-parameter λ .

	AP^{val}_{75}	AP^{val}_{50}	AP^{val}_{50}
PIoU v1(baseline)	54.12	49.72	69.26
PIoU v2($\lambda = 1.1$)	53.99(−0.13)	49.89(+0.17)	69.31(+0.05)
PIoU v2($\lambda = 1.2$)	54.62(+0.50)	50.23(+0.51)	69.58(+0.32)
PIoU v2($\lambda = 1.3$)	54.45(+0.33)	50.43(+0.71)	69.96(+0.70)
PIoU v2($\lambda = 1.4$)	53.51(−0.61)	49.81(+0.09)	69.42(+0.16)
PIoU v2($\lambda = 1.5$)	54.67(+0.55)	50.29(+0.57)	69.95(+0.69)
PIoU v2($\lambda = 1.6$)	54.05(−0.07)	49.87(+0.15)	69.67(+0.41)
PIoU v2($\lambda = 1.7$)	54.15(+0.03)	49.91(+0.19)	69.81(+0.55)

Table 2. PIoU v2 achieved state-of-the-art (SOTA) accuracy among all loss functions, with gains of 0.5% AP_{75} , 0.3% AP . Compared to DINO's (Rezatofighi et al., 2019) original GIoU (Rezatofighi et al., 2019) loss function, PIoU v2 achieved gains of 1.1% AP_{75} , 0.6% AP .

YOLOv8 on IC SEM dataset. We present the comparison results with other existing loss functions on the IC SEM dataset in Table 3. The model using PIoU v2 achieved the highest accuracy, demonstrating the superiority of PIoU v2.

Compared to EIoU (Zhang, Ren et al., 2022), Focal-EIoU showed a decrease in accuracy, indicating the instability of the focusing mechanism in Focal-EIoU. The enhancements of PIoU v2 over PIoU v1 are more stable.

5.3. Ablation experiments

Hyper-parameter. Table 4 shows the effect of the hyperparameter λ on the performance of the model using PIoU v2. We observed that PIoU v2 is sensitive to the variations of the hyper-parameter λ and works best when the hyper-parameter $\lambda = 1.3$. When λ is larger or smaller, the gradient gain of high-quality samples and the P corresponding to the peak of the gradient become inappropriate for convergence.

Function form. We compared the impact of different functional forms with the same penalty factor on the model performance. The results in Table 5 show that the function we used, $f(x)$, achieves the best performance among the three functions.

The function $f(x)$ assigns a large gradient to speed up the regression of the anchor box when the anchor box quality is medium, and the gradient is reduced appropriately when the anchor box is of high quality in order to optimize the anchor box steadily further toward the target box. Moreover, when the anchor box is of inferior quality, its gradient is harmful, and a smaller gradient is given to restrain the harmful



Fig. 8. Detection examples using YOLOv8 m trained on MS COCO dataset. Left to right: PIoU v2, Focal-EIoU, WIoU.

gradient from low-quality anchor boxes. Our function $f(x)$ possesses these properties, implying that it achieves adaptive gradient adjustment for the anchor box quality without additional weights, corresponding to the non-monotonic focusing mechanism in WIoU (Tong et al., 2023), facilitating the performance improvement of object detectors. The difference is that the focusing mechanism of PIoU is static. In contrast, the function $g_1(x)$ gradient increases as the anchor box quality worsens, bringing harmful gradients from low-quality anchor boxes. $g_2(x)$ has a constant gradient and lacks attention to the anchor box quality.

The experimental results demonstrate that our function $f(x)$ helps object detectors achieve the best performance among the three functions. Further, for possible special cases such as the initial anchor boxes are all of very low quality, there are probably better functions that can be combined with the penalty factor P to form more effective loss functions, and we hope future work will carry out more research.

Table 5

Performance of different functional forms.

Loss/ Evaluation	AP_{75}^{val}	AP^{val}	AP_{50}^{val}
L_{PIOU}	54.12	49.72	69.26
L_1	54.02	49.71	68.91
L_2	53.53	49.50	68.70

6. Conclusion

In this paper, we have analyzed the reasons for anchor box enlargement and other limitations of existing IoU-based loss functions. We propose PIoU loss with more powerful penalty factors and a better functional form to tackle these drawbacks, achieving faster convergence and higher accuracy. Further, we designed a version with enhanced

focusing capability, PIoU v2, to be incorporated into popular object detectors. After comparison and ablation experiments, our proposed PIoU v2 achieves the highest accuracy on the COCO, VOC and IC SEM datasets, verifying the effectiveness of the innovation point proposed in this paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The code will be available as: <https://github.com/fppccc/Powerful-IOU>.

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